LUMPING AND THE MIXING TIME OF A REVERSIBLE CHAIN: THE DEPENDENCE ON THE NUMBER OF STATES

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Abstract. Aldous and Fill asked whether the stopping-rule mixing time $\tau_1^{(2)}$ of a chain obtained from a reversible Markov chain by lumping states is at most an absolute constant times that of the original chain. Hermon and Peres answered the question in the negative: for random walks on graphs of bounded degree, lumping can increase the mixing time by a factor of the order of the logarithm of the number of vertices, and by no more. We consider general reversible chains with n states and show that the largest possible factor $K(n)$ satisfies $\lfloor (n-2)/2 \rfloor \le K(n) \le n-1$. The upper bound follows from standard inequalities and from the inequality $\tau_2 \le \tau_1^{(2)}$ between the relaxation time and the stopping-rule mixing time, for which we give a short proof; the lower bound comes from an explicit random walk on a weighted comb. We also show that lumping increases the mean hitting time of a set A by a factor of at most $n - |A|$, and that this bound cannot be improved by more than a factor of about two.

1. Introduction

Let I be a finite set with $n \ge 2$ elements and let $(w_{xy})_{x,y \in I}$ be a symmetric array of non-negative weights, loops $w_{xx} > 0$ being allowed, such that the graph with edges $\{x, y\}$, $w_{xy} > 0$, is connected. Put $w_x = \sum_y w_{xy}$ and $w = \sum_x w_x$. The random walk on this network is the Markov chain X on I with transition probabilities $p_{xy} = w_{xy}/w_x$; it is reversible with stationary distribution $\pi_x = w_x/w$, and every irreducible reversible chain on I is of this form. We write $T_y = \inf\{t \ge 0 : X_t = y\}$ for the hitting time of a state y, $T_A = \inf\{t \ge 0 : X_t \in A\}$ for the hitting time of a set A, so that $T_y = 0$ for the walk started at y, and $\mathbb{E}_\mu$ for expectation for the chain started from the distribution μ.

Given a map f from I onto a set $\widehat{I}$, the *induced chain* $\widehat{X}$ is the random walk on $\widehat{I}$ with weights

$$\widehat{w}_{ab} = \sum_{x \in f^{-1}(a)} \sum_{y \in f^{-1}(b)} w_{xy}, \qquad a, b \in \widehat{I}.$$

Its transition probabilities are $\mathbb{P}_\pi(f(X_1) = b \mid f(X_0) = a)$ and its stationary distribution is $\widehat{\pi}_a = \pi(f^{-1}(a))$; this is the chain of Aldous and Fill [2, Section 4.6.1]. In the language of networks, the vertices with the same value of f are glued together. We say that $\widehat{X}$ is obtained from X by lumping, and write $\widehat{\mathbb{E}}$ and $\widehat{T}$ for its expectations and hitting times.

The parameter of [2, Chapter 4] that we study is

$$\tau_1^{(2)} = \max_{i \in I} \min_U \mathbb{E}_i U,$$

where the minimum is over all stopping times U, possibly randomized with a randomization independent of the future of the chain, such that X_U has distribution π under $\mathbb{P}_i$. Lovász and Winkler [4] showed that the minimum is attained and found its value (see also [2, Theorem 9.4]),

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so that

$$\tau_1^{(2)} = \max_{i,j\in I}\bigl(\mathbb{E}_i T_j - \mathbb{E}_\pi T_j\bigr). \tag{1}$$

For reversible chains $\tau_1^{(2)}$ is equivalent, up to absolute constants, to the total variation mixing time of the chain in continuous time [1], [2, Theorem 4.6]; see also [5].

The maximal mean commute time $\tau^* = \max_{i,j}(\mathbb{E}_i T_j + \mathbb{E}_j T_i)$, the average hitting time $\tau_0 = \sum_j \pi_j \mathbb{E}_i T_j$ (which does not depend on i), the relaxation time $\tau_2 = 1/(1-\lambda_2)$, where λ_2 is the second largest eigenvalue of (p_{xy}), and the Cheeger time constant τ_c do not increase under lumping [2, Proposition 4.44]. For the mixing time the matter was left open:

Open Problem 4.45 of [2]**.** *Is the value of $\tau_1^{(2)}$ in an induced chain bounded by K times the value of $\tau_1^{(2)}$ in the original chain, for some absolute constant K? For $K = 1$?*

Hermon and Peres [3, Theorem 3(c)] answered both questions in the negative. They construct graphs G_n of bounded degree for which lumping together certain pairs of vertices increases the total variation mixing time of the lazy simple random walk by a factor of order $\log|V(G_n)|$. Lumping does not increase the relaxation time, and the mixing time of a lazy reversible chain is bounded below by a constant multiple of the relaxation time and above by a constant multiple of the relaxation time multiplied by $\log(1/\min_x \pi_x)$; hence this is the largest possible order in that setting [3, Remark 1.8].

For a random walk on a weighted network $\min_x \pi_x$ can be arbitrarily small, and the bound in terms of $\log(1/\min_x \pi_x)$ says nothing about the dependence on the number of states. This note determines that dependence up to a factor of about two. Let

$$K(n) = \sup \ \frac{\tau_1^{(2)}(\widehat{X})}{\tau_1^{(2)}(X)},$$

the supremum being over all reversible chains X with at least two and at most n states and all induced chains $\widehat{X}$ of X.

Theorem 1. *For every $n \geq 2$,*

$$\max\Bigl(1, \Bigl\lfloor \frac{n-2}{2} \Bigr\rfloor\Bigr) \ \leq \ K(n) \ \leq \ n-1.$$

The upper bound is a chain of four inequalities, three of which are in [2]. The fourth is the following comparison of the relaxation time with the stopping-rule mixing time, in which the constant is one; whether this holds is asked in [2, Open Problem 4.48(b)]. Comparisons of the two quantities up to absolute constants are in [1, 5].

Lemma 2. *Let $\lambda < 1$ be a real eigenvalue of the transition matrix of an irreducible chain on a finite set, with a real eigenfunction g, and let i be a state at which $|g|$ is largest. If U is a stopping time such that X_U has the stationary distribution π under $\mathbb{P}_i$, then $\mathbb{E}_i U \geq 1/(1-\lambda)$. In particular $\tau_2 \leq \tau_1^{(2)}$ for every reversible chain.*

Equality $\tau_2 = \tau_1^{(2)}$ holds for every chain with two states and for the simple random walk on a complete graph.

The lower bound in Theorem 1 is approached along a family of random walks on a weighted comb with $2N + 2$ states, described in Section 3; lumping each tooth of the comb with a vertex of the spine multiplies $\tau_1^{(2)}$ by a factor that can be made arbitrarily close to N. In these examples the stationary distribution is very far from uniform, in agreement with the bound of [3]; see Remark 4.

The mechanism is a one-sided hitting time. Aldous and Fill observe [2, Section 4.6.1] that, although commute times decrease under lumping, a mean hitting time may increase, and give an example on four vertices where it doubles. The next proposition bounds the increase for hitting times of sets.

Proposition 3. *Let $\widehat{X}$ be the chain induced from X by $f : I \to \widehat{I}$, let $\widehat{A}$ be a non-empty proper subset of $\widehat{I}$ and $A = f^{-1}(\widehat{A})$. Then for every $a \in \widehat{I} \setminus \widehat{A}$*

$$\widehat{\mathbb{E}}_a \widehat{T}_{\widehat{A}} \;\le\; \sum_{y \in I \setminus A} \mathbb{E}_y T_A \;\le\; (n - |A|) \max_{y \in I} \mathbb{E}_y T_A.$$

For every $N \ge 1$ and $\varepsilon > 0$ there is a chain with $n = 2N + 2$ states, an induced chain and a set A with one element such that $\widehat{\mathbb{E}}_a \widehat{T}_{\widehat{A}} \ge (N - \varepsilon) \max_y \mathbb{E}_y T_A$ for some a.

Lemma 2 and the upper bound of Theorem 1 are proved in Section 2, the lower bound in Section 3 and Proposition 3 in Section 4. We do not know whether $K(n)/n$ converges, nor the value of $K(4)$; an example with four states and ratio above 1.27 is in Remark 7.

2. The upper bound

Proof of Lemma 2. Replacing g by a multiple we may assume that $g(i) = 1 = \max_x |g(x)|$. Since $Pg = \lambda g$, the process

$$M_t = g(X_t) + (1 - \lambda) \sum_{s<t} g(X_s), \qquad t \ge 0,$$

is a martingale with $M_0 = 1$ under $\mathbb{P}_i$, with respect to the filtration generated by the chain and by the randomization used by U. We may assume that $\mathbb{E}_i U < \infty$. The increments of M are bounded by 2, so the optional stopping theorem gives $\mathbb{E}_i M_U = 1$. As $\lambda \ne 1$, the function g is orthogonal to the constants in $L^2(\pi)$ when the chain is reversible, and in general $\sum_x \pi_x g(x) = \sum_x \pi_x (Pg)(x) = \lambda \sum_x \pi_x g(x)$ forces $\sum_x \pi_x g(x) = 0$. Hence $\mathbb{E}_i g(X_U) = 0$ and

$$1 = (1 - \lambda)\, \mathbb{E}_i \sum_{s<U} g(X_s) \le (1 - \lambda)\, \mathbb{E}_i U.$$

For a reversible chain all eigenvalues are real; taking $\lambda = \lambda_2$ gives $\tau_1^{(2)} \ge \min_U \mathbb{E}_i U \ge \tau_2$. □

Proof of the upper bound in Theorem 1. Let X have $m \le n$ states and let $\widehat{X}$ be an induced chain. We claim that

$$\tau_1^{(2)}(\widehat{X}) \;\le\; \tau_0(\widehat{X}) \;\le\; \tau_0(X) \;\le\; (m-1)\, \tau_2(X) \;\le\; (m-1)\, \tau_1^{(2)}(X). \tag{2}$$

For the first inequality, choose a state J of $\widehat{X}$ with distribution $\widehat{\pi}$, independently of the chain, and stop at the hitting time of J; the chain is then in the stationary distribution, and the mean of this stopping time is $\tau_0(\widehat{X})$ from every starting state [2, Section 4.3]. The second inequality is part of the contraction principle [2, Proposition 4.44]: by the random target lemma $\tau_0 = \frac{1}{2} \sum_{x,y} \pi_x \pi_y (\mathbb{E}_x T_y + \mathbb{E}_y T_x)$, the commute time $\mathbb{E}_x T_y + \mathbb{E}_y T_x$ equals w times the effective resistance between x and y, and gluing vertices does not increase effective resistances. The third is the eigentime identity $\tau_0 = \sum_{k \ge 2} (1 - \lambda_k)^{-1}$ [2, Proposition 3.13 and Lemma 4.24]. The fourth is Lemma 2. If $\widehat{X}$ has one state then $\tau_1^{(2)}(\widehat{X}) = 0$, and $\tau_1^{(2)}(X) \ge \tau_2(X) \ge \frac{1}{2}$, so the ratio is well defined; hence $K(n) \le n - 1$. □

Remark 4. Write $\pi_* = \min_x \pi_x$. In discrete time $\tau_1^{(2)} \le \frac{4e}{e-1}\,\tau_2\,(1+\frac{1}{2}\log(1/\pi_*))$ by [2, Lemma 4.23]. Since $\tau_2(\widehat{X}) \le \tau_2(X)$ and $\min_a \widehat{\pi}_a \ge \pi_*$, Lemma 2 gives

$$\tau_1^{(2)}(\widehat{X}) \;\le\; \frac{4e}{e-1}\Big(1+\frac{1}{2}\log\frac{1}{\pi_*}\Big)\tau_1^{(2)}(X),$$

which is the form taken for $\tau_1^{(2)}$ by the bound of Hermon and Peres [3, Remark 1.8]. Together with Theorem 1, the factor is at most $\min\big(n-1,\ \frac{4e}{e-1}(1+\frac{1}{2}\log(1/\pi_*))\big)$. For the simple random walk on a graph with bounded degrees $\log(1/\pi_*)$ is of order $\log n$, and the examples of [3] show that this order is attained. In the comb of Section 3 with the parameters of Table 1, $\log(1/\pi_*)$ is of order $N\log t$.

3. The comb

We use the following elementary fact about hitting times across a bridge.

Lemma 5. *Let $x \ne y$ be states with $w_{xy} > 0$ such that every path from x to y in the network uses the edge $\{x,y\}$, and let S be the set of states that can be reached from x without visiting y. Then $\mathbb{E}_x T_y = w(S)/w_{xy}$, where $w(S) = \sum_{u\in S} w_u$.*

Proof. Up to time T_y the walk started at x stays in S, so $\mathbb{E}_x T_y$ is the same for the network on $S \cup \{y\}$ in which y keeps only the edge to x. There the stationary probability of y is $w_{xy}/(w(S)+w_{xy})$, so the mean return time to y is $1+w(S)/w_{xy}$, and it also equals $1+\mathbb{E}_x T_y$. □

Fix an integer $N \ge 1$ and real numbers $r \ge 2$, $\alpha > 0$, $h \ge 1$ and $\ell \ge 0$. The *comb* is the network on the $2N+2$ states $z_0,\dots,z_N$, $y_0,\dots,y_{N-1}$, c with the weights

$$w_{z_k z_{k+1}} = r^k, \quad w_{y_k z_{k+1}} = \alpha r^k, \quad w_{y_k y_k} = (h-1)\alpha r^k \quad (0 \le k < N), \qquad w_{z_N c} = r^N, \quad w_{cc} = \ell,$$

and all other weights zero. The spine $z_0,\dots,z_N,c$ carries conductances that grow geometrically towards c; the tooth y_k hangs from z_{k+1}, and the walk leaves it with probability $1/h$ at each step. The map f sends z_k and y_k to the same point b_k for $0 \le k < N$, sends z_N to b_N and c to c. Put

$$\delta = \frac{2+\alpha+\alpha h}{r-1}, \qquad \rho = \frac{w_c}{r^N} = 1+\frac{\ell}{r^N}, \qquad M = h+(N+1)(1+\delta), \qquad B = h+(N+2)(1+\delta).$$

Proposition 6. *For the comb X and its induced chain $\widehat{X}$,*

(i) $\max_x \mathbb{E}_x T_c \le M$;
(ii) $\tau_1^{(2)}(X) \le \max(B,\ M+B-\rho)$, *so that* $\tau_1^{(2)}(X) \le B$ *when* $\rho \ge M$;
(iii) $\widehat{\mathbb{E}}_{b_0}\widehat{T}_c \ge Nh\alpha/(1+\alpha)$ *and* $\tau_1^{(2)}(\widehat{X}) \ge \pi_c\, Nh\alpha/(1+\alpha)$.

Proof. The comb is a tree with loops. The vertex weights are $w_{z_0} = 1$, $w_{z_k} = r^{k-1}(1+r+\alpha)$ for $1 \le k \le N$ and $w_{y_k} = h\alpha r^k$. Write $z_{N+1} = c$. For $0 \le k \le N$ let S_k be the total weight of the states $z_0,\dots,z_k,y_0,\dots,y_{k-1}$, which are those separated from c by the edge $\{z_k,z_{k+1}\}$. Since $(1+r+\alpha+h\alpha)/(r-1) = 1+\delta$,

$$S_k = 1+(1+\delta)(r^k-1) \le (1+\delta)\,r^k. \tag{3}$$

(i) By Lemma 5, $\mathbb{E}_{z_k} T_{z_{k+1}} = S_k/r^k \le 1+\delta$ and $\mathbb{E}_{y_k} T_{z_{k+1}} = h$. Adding along the path to c gives $\mathbb{E}_{z_k} T_c \le (N+1)(1+\delta)$ and $\mathbb{E}_{y_k} T_c \le h + N(1+\delta)$.

(ii) Write $H(x,j) = \mathbb{E}_x T_j$ and $d(x,j) = H(x,j) - \mathbb{E}_\pi T_j = \sum_k \pi_k\big(H(x,j)-H(k,j)\big)$; by (1), $\tau_1^{(2)}(X) = \max_{x,j} d(x,j)$. For three vertices x,k,j of a tree let $m(x,k,j)$ be the vertex common

to the three paths joining them. A walk from x to j passes through $m = m(x,k,j)$, so $H(x,j) = H(x,m) + H(m,j)$, and likewise from k; hence

$$H(x,j) - H(k,j) = H(x,m) - H(k,m) \le H(x,m). \tag{4}$$

If $j = c$ then $d(x,c) \le H(x,c) \le M$ by (i).

Let $x = c$ and $j \ne c$. Define the level of z_p to be p and the level of y_p to be $p+1$, and let q be the level of j. By Lemma 5, $\mathbb{E}_{z_{t+1}} T_{z_t} = (w - S_t)/r^t \le w\, r^{-t}$, so for $0 \le p \le N$

$$H(c, z_p) \le w \sum_{t \ge p} r^{-t} = \frac{r}{r-1}\, w\, r^{-p} \le 2w\, r^{-p}, \tag{5}$$

and $\mathbb{E}_{z_{p+1}} T_{y_p} = (w - w_{y_p})/(\alpha r^p) \le w/(\alpha r^p)$. By (4), $d(c,j) \le \sum_{k \ne c} \pi_k H(c, m(c,k,j))$. If the level p of k exceeds q then $m(c,k,j) = z_p$; the two states of level p have total weight $r^{p-1}(1 + r + \alpha + h\alpha)$, so by the geometric series in (5) they contribute at most $(1 + r + \alpha + h\alpha)/(r-1) = 1 + \delta$, and there are $N - q$ such levels. If the level of k is at most q and $k \ne j$ then $m(c,k,j) = z_q$; the same holds for $k = j$ when $j = z_q$, while for $k = j = y_{q-1}$ the median is j and $H(c,j) = H(c,z_q) + H(z_q, y_{q-1})$. The states of level at most q, with j among them, have total weight S_q, so by (3) and (5) the terms $\pi_k H(c, z_q)$ contribute at most $2(1+\delta)$. If $j = y_{q-1}$ there is in addition the term $\pi_j H(z_q, y_{q-1})$, which is at most $w_{y_{q-1}}/(\alpha r^{q-1}) = h$. Altogether

$$d(c,j) \le h + (N - q + 2)(1 + \delta) \le B.$$

Let $x \ne c$ and $j \ne c$. Since c is a leaf, $m = m(x,c,j) \ne c$, and m lies on the path from x to c and on the path from c to j. By (4), (i) and Lemma 5,

$$H(x,j) - H(c,j) = H(x,m) - H(c,m) \le H(x,c) - H(c, z_N) \le M - \rho,$$

so $d(x,j) = H(x,j) - H(c,j) + d(c,j) \le M - \rho + B$. As $d(c,c) \le 0$ and $M \le B$, this proves (ii).

(iii) The induced chain is a random walk on the path $b_0, \dots, b_N, c$ with $\widehat{w}_{b_k b_{k+1}} = (1+\alpha) r^k$ for $k < N$, $\widehat{w}_{b_N c} = r^N$ and loops. By Lemma 5, for $k < N$

$$\widehat{\mathbb{E}}_{b_k} \widehat{T}_{b_{k+1}} \;\ge\; \frac{\widehat{w}_{b_k}}{(1+\alpha) r^k} \;\ge\; \frac{w_{y_k}}{(1+\alpha) r^k} = \frac{h\alpha}{1+\alpha},$$

and the sum over k gives the first assertion. On a path $\widehat{\mathbb{E}}_{b_0} \widehat{T}_c \ge \widehat{\mathbb{E}}_a \widehat{T}_c$ for every a, so by (1)

$$\tau_1^{(2)}(\widehat{X}) \;\ge\; \sum_a \widehat{\pi}_a \big(\widehat{\mathbb{E}}_{b_0} \widehat{T}_c - \widehat{\mathbb{E}}_a \widehat{T}_c \big) \;\ge\; \widehat{\pi}_c\, \widehat{\mathbb{E}}_{b_0} \widehat{T}_c,$$

and $\widehat{\pi}_c = \pi_c$. □

For the parameters used below, where r is much larger than αh, a walk on the spine of the comb rarely enters a tooth, and the teeth cost a walk about one sojourn, of mean h, on its way to c. After lumping, every point of the spine below z_N inherits the loop of a tooth, and when α is large the delay is about h at each of the N levels.

Proof of the lower bound in Theorem 1. The identity map gives $K(n) \ge 1$. Let $N \ge 1$ and $t \ge 2$, and take $\alpha = h = t$, $r = 1 + t^3$ and ℓ so large that $\rho \ge M$ and $\pi_c \ge 1 - 1/t$. Then $\delta = (2 + t + t^2)/t^3 \le 1$ and, by Proposition 6,

$$\frac{\tau_1^{(2)}(\widehat{X})}{\tau_1^{(2)}(X)} \;\ge\; \Big(1 - \frac{1}{t}\Big) \frac{N\, t^2}{(1+t)\big(t + (N+2)(1+\delta)\big)},$$

which tends to N as $t \to \infty$. The comb has $2N + 2$ states, so $K(n) \ge N$ whenever $n \ge 2N + 2$. □

Table 1 gives the values for $\alpha = h = t = 256$, $r = 1 + t^3$ and $w_c = (N+2)t^2(w - w_c)$, computed in rational arithmetic from (1) and rounded.

N	states	$\tau_1^{(2)}(X)$	$\tau_1^{(2)}(\widehat{X})$	ratio
1	4	257.004	256.012	0.9961
2	6	258.008	511.019	1.9806
4	10	260.016	1021.035	3.9268
8	18	264.031	2041.066	7.7304

TABLE 1. The comb with $\alpha = h = 256$ and its induced chain.

Remark 7. The comb with $N = 1$ has four states, and with the parameters of Table 1 its ratio is below one. A different chain on four states has ratio above one: take $I = \{0,1,2,3\}$ with $w_{02} = w_{12} = 1$, $w_{13} = 109$, $w_{23} = 54$, $w_{11} = 117$, $w_{33} = 10000$, and lump the states 2 and 3. Then

$$\tau_1^{(2)}(X) = \frac{149731115}{63193903} = 2.3693\ldots, \qquad \tau_1^{(2)}(\widehat{X}) = \frac{346900}{114917} = 3.0187\ldots,$$

both attained in (1) at $(i,j) = (1,0)$, and the ratio is $419679620/329408453 = 1.2740\ldots$. Hence $K(4) > 1.27$.

4. Hitting times of sets

For a non-empty proper subset Z of I and $x, y \notin Z$ let $G_Z(x,y)$ be the expected number of visits to y before T_Z for the walk started at x, and let $R(y \leftrightarrow Z)$ be the effective resistance between y and Z in the network with conductances w_{uv}, $u \neq v$. Then [6, Section 2.2]

(6)
$$\mathbb{E}_x T_Z = \sum_{y \notin Z} G_Z(x,y), \qquad G_Z(x,y) = \mathbb{P}_x(T_y < T_Z)\, G_Z(y,y), \qquad G_Z(y,y) = w_y\, R(y \leftrightarrow Z).$$

The last identity holds in the presence of loops: the walk at y reaches Z before returning to y with probability $1/(w_y R(y \leftrightarrow Z))$, and a loop at y changes w_y only.

Proof of Proposition 3. The network of $\widehat{X}$ is obtained from that of X by gluing together the vertices in each set $f^{-1}(b)$; edges inside such a set become loops. By Rayleigh's monotonicity principle [6, Section 2.4] gluing does not increase effective resistances, so $\widehat{R}(b \leftrightarrow \widehat{A}) \le R(y \leftrightarrow A)$ for every $b \in \widehat{I} \setminus \widehat{A}$ and every $y \in f^{-1}(b)$. With (6) applied to both chains,

$$\begin{aligned} \widehat{\mathbb{E}}_a \widehat{T}_{\widehat{A}} \;&\le\; \sum_{b \notin \widehat{A}} \widehat{w}_b\, \widehat{R}(b \leftrightarrow \widehat{A}) = \sum_{b \notin \widehat{A}} \sum_{y \in f^{-1}(b)} w_y\, \widehat{R}(b \leftrightarrow \widehat{A}) \\ &\le\; \sum_{y \notin A} w_y\, R(y \leftrightarrow A) = \sum_{y \notin A} G_A(y,y) \;\le\; \sum_{y \notin A} \mathbb{E}_y T_A. \end{aligned}$$

For the second statement take the comb with $A = \{c\}$, $\widehat{A} = \{c\}$ and $a = b_0$. By Proposition 6(i) and (iii),

$$\frac{\widehat{\mathbb{E}}_{b_0} \widehat{T}_c}{\max_y \mathbb{E}_y T_c} \;\ge\; \frac{Nh\alpha}{(1+\alpha)\big(h + (N+1)(1+\delta)\big)},$$

and with $\alpha = h = t$, $r = 1 + t^3$ the right-hand side tends to N as $t \to \infty$. □

The program in the repository named below evaluates both sides of every inequality of Proposition 6 in rational arithmetic for $N \le 8$ and $\alpha = h = t \le 256$, and does the same for (2) with τ_2 omitted and for Proposition 3 on 400 random networks with three to six states and random partitions.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [9] and Math Brain [7]. In October 2026 an AI agent connected to the Brain, working from the project's list of open problems in probability, found examples in which lumping increases $\tau_1^{(2)}$, among them the example of Remark 7. In November 2026 a Claude Fable agent (Anthropic) obtained Theorem 1, Lemma 2 and Propositions 3 and 6 with their proofs, wrote and ran the programs and prepared the manuscript. A Codex agent (OpenAI) then read the proofs and recomputed the examples with programs of its own. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/lumping-mixing-time`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [7].

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